KUSH MEHTA

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Final-year Electronics & Communication Engineering student working in digital design and computer architecture. Built a synthesizable GF(2^8) arithmetic core in Verilog, verified against golden vectors from C++ written during a research internship at ISRO, where I implemented Reed–Solomon erasure decoding for a real-time GNSS correction pipeline.

EDUCATION

Institute of Technology, Nirma University — Ahmedabad, India
B.Tech, Electronics & Communication Engineering — Minor: Data Science | CGPA: 8.59 / 10 — Aug 2023 – May 2027

- **Coursework:** Embedded Systems, Computer Architecture, Microcontrollers & Interfacing, Scripting Languages, VLSI Design, Digital Signal Processing, Programming for Scientific Computing

EXPERIENCE

Research Intern — Navigation Receiver Division, Space Applications Centre (SAC), ISRO — May – Jul 2026
Real-Time Galileo High Accuracy Service (HAS) Decoding Utility · 10-week research internship — Ahmedabad, India

- Built a real-time C++ utility that ingests a multiplexed TCP stream, tokenizes three interleaved protocols (RTCM payloads, Galileo HAS E6-B pages, Time-of-Transfer epoch tags), and routes each to independent output ports.
- Implemented Reed–Solomon (255, 32) erasure decoding over GF(2^8) from the Galileo HAS ICD — lookup-table field arithmetic and Gaussian elimination for k×k matrix inversion — recovering complete messages despite page loss and out-of-order arrival.
- Reverse-engineered four undocumented Galileo RTCM 3.x message formats (1240–1243) and the CRC-24Q framing routine by cross-referencing an open-source implementation against the ICD.
- Delivered streaming over WinSock 2.2 TCP and Win32 serial with background-thread reads and stale-buffer purging, working around a pre-C++11 toolchain that ruled out modern STL containers and threading.
- Validated end-to-end in RTKLIB/RTKNAVI at centimetre-level Precise Point Positioning, with each decoding stage unit-verified against a Python reference implementation.

Electronics Lead — Team Arrow UAV Club, Nirma University — Sep 2023 – Present
Official Aeromodelling & Autonomous Systems Club — Ahmedabad, India

- Own the avionics and embedded stack for competition aircraft: flight-controller configuration (ArduPilot / Pixhawk), telemetry links, sensor integration, ESC and motor selection, and power system design.
- Designed autonomous payload electronics — magnetometer, time-of-flight ranging, IR line-following — for a ground vehicle deployed from a tilt-rotor VTOL aircraft; validated the mission in software-in-the-loop before hardware.
- Lead a team of 8–12 and mentor juniors on firmware, sensor fusion, and avionics integration. Four podium finishes across national and international SAE and Technoxian competitions, including AIR 2 (SAE DDC Micro 2024) and 5th internationally in mission score (SAE Aero Design West 2025, USA).

PROJECTS

GF(2^8) Arithmetic Core in Verilog — 2026
github.com/Kush-Git-379/gf256-rtl

- Rebuilt the Galois-field arithmetic from the ISRO Reed–Solomon erasure decoder as synthesizable RTL over primitive polynomial 0x11D: two GF(2^8) multiplier microarchitectures (log/antilog ROM and combinational shift-and-reduce), table inversion, and multiply-accumulate.
- Wrote self-checking SystemVerilog testbenches driven by golden vectors from the C++ original, covering all 65,536 multiplier pairs and 256 inverse cases. Proved the tests could fail by injecting three seeded faults.
- Quartus synthesis on Cyclone IV E made shift-and-reduce 13× smaller and 1.7× faster than the table version (62 LEs at 7.71 ns against 804 at 12.95 ns). Traced it to asynchronous ROM reads blocking M9K inference; registering the read address recovered block RAM and raised F_{max} to 180 MHz.

Secure GPS Telemetry Pipeline — ESP32, AES-128, MQTT — Mar 2026

- End-to-end encrypted IoT pipeline: ESP32 parses NMEA over UART2 with TinyGPS++, serializes to JSON, encrypts with AES-128 via mbedTLS, and publishes hex ciphertext to an MQTT broker on a 3-second cycle.
- Python subscriber (paho-mqtt, PyCryptodome) decrypts and resolves the live position; verified on hardware from GPS fix through encrypted transport to decoded coordinates.

RISC-V Cache Locality: Same Instructions, 11.8× Fewer Misses — 2026

- Wrote pairs of RV32IM programs executing identical instructions and memory operations, differing only in access order, and measured each in Ripes on a 5-stage RV32IMC core (4 KiB, 2-way, LRU, write-back). Loop order alone moved the hit rate 0%→75% on the same 32,967 instructions and 4,096 reads; on a 64×64 matmul, padding the row stride 64→65 words then blocking took the miss rate 51.5%→13.2%→4.3%, an 11.8× reduction.
- Tested whether hardware could substitute for code: 2-way→4-way associativity changed nothing (48.5% either way) and only a fully associative 256-way cache reached 99.1%, isolating the power-of-two stride as the cause rather than capacity.

Cryptography Benchmark for Resource-Constrained Devices — Jan 2026

- Benchmarked AES-GCM, DES, ECC P-256, and ASCON-128 across 5,000 IoT payloads; loaded the ASCON C reference as a compiled DLL through Python `ctypes` for native-speed timing, profiling peak memory with `tracemalloc`. ASCON ran 78× faster than AES-GCM and 12× faster than DES.

TECHNICAL SKILLS

Languages: C, C++, Verilog HDL, SystemVerilog, Python, Assembly, TCL, Bash, MATLAB, SQL

Digital Design & Verification: RTL design, functional verification, testbench development, golden-reference testing, fault injection, logic synthesis, static timing analysis, ModelSim, Quartus II, FPGA

Computer Architecture: RISC-V RV32IM, gem5, microarchitecture, cache and memory hierarchy, set associativity, IPC and performance analysis, workload characterization, hardware/software co-design

Analog & Circuit Design: Cadence Virtuoso, LTspice, SPICE simulation, transient / DC / AC analysis, pre- and post-layout simulation, MOSFET parameter extraction, cascode and differential amplifier design

Embedded & Firmware: ESP32, 8051, Arduino, Pixhawk, Keil uVision, mbedTLS, UART / I^2C / SPI, real-time systems

Algorithms & Standards: Reed–Solomon, forward error correction, Galois Field GF(256), RTCM 3.x, CRC-24Q, AES-128, ASCON-128, DSP

Tools: Git / GitHub, Linux, RTKLIB, pandas, NumPy, OpenCV